

Project 2

Smart City Mobility & Air Quality Analytics (Excel Simulation Project)

Excel Statistical Analysis Project

1. Project Overview

In this project, you will analyse a **1,000-row Smart City dataset** containing information about:

- Transport modes
- Journey distances and durations
- Delays and incidents
- Weather conditions
- Air quality
- Passenger satisfaction
- CO₂ emissions
- Travel patterns across the city

You will use Excel to explore:

- Filtering
- Descriptive statistics
- Correlation
- Linear regression
- Logistic regression
- Covariance
- Visualisation techniques (bar, pie, line, scatter, 2D/3D)

You will finish with a **professional report** suitable for your portfolio.

2. Dataset Description

You will be provided with a **1,000-row synthetic dataset** containing the following fields:

- Journey_ID
- Day_of_Week
- Time_of_Day
- Transport_Mode (car, bus, train, bike, walk, e-scooter)
- Distance_km
- Journey_Duration_mins
- Average_Speed
- Delay_mins
- Weather_Type

- Temperature_C
- Air_Quality_Index
- CO2_Emissions_g
- Passenger_Count
- Ticket_Price
- Satisfaction_Score (1–10)
- Incident_Reported (Yes/No)
- Late_Arrival (Yes/No)
- Frequent_Traveller (Yes/No)

You will receive the dataset as an **Excel file**.

3. Step 1 — Data Familiarisation

Open the dataset and explore:

- Number of rows
- Number of columns
- Types of variables
- Any missing or unusual values
- Any patterns you notice at first glance

Write a short **initial observations** section.

4. Step 2 — Filtering (Excel)

Apply filters to explore:

- Transport mode
- Day of week
- Weather type
- Delay ranges
- Air quality levels
- Satisfaction score groups

You must produce:

- **5 filtered views**
- A short explanation of what you noticed in each view

Visuals required:

- 1 bar chart
- 1 pie chart
- 1 conditional-formatted table

5. Step 3 — Descriptive Statistics

Calculate the following for at least **five numeric fields**:

- Mean
- Median
- Mode
- Range
- Standard deviation
- Variance
- Min/Max
- Quartiles

Suggested fields:

- Distance
- Duration
- Delay
- Air Quality Index
- Satisfaction Score
- CO₂ Emissions

Visuals required:

- Box-and-whisker plot
- Histogram
- 2D or 3D column chart

Write a short explanation of **patterns and trends**.

6. Step 4 — Correlation Analysis

Calculate correlations between meaningful pairs, such as:

- Distance vs Duration
- Speed vs Delay
- AQI vs Temperature
- Delay vs Satisfaction
- Passenger Count vs Ticket Price

Visuals required:

- Scatter graphs (with trendlines)
- Correlation table (colour-coded)

Explain which relationships are strong, weak, or none.

7. Step 5 — Linear Regression

Run **two** linear regressions using Excel's Data Analysis Toolpak:

1. Distance_km → Journey_Duration_mins
2. Delay_mins → Satisfaction_Score

Interpret:

- R^2
- Coefficients
- Significance
- Residuals

Visuals required:

- Scatter graphs with regression lines

Explain what the regression predicts and how accurate it is.

8. Step 6 — Logistic Regression

Use a binary variable such as:

- **Late_Arrival (Yes/No)**
or
- **Incident_Reported (Yes/No)**

Predictors may include:

- Delay
- Weather
- Transport mode (converted to numeric)
- Air Quality Index
- Time of day

Steps:

- Convert Yes/No → 1/0
- Run logistic regression (Solver or manual method)
- Interpret:
 - Odds ratios
 - Coefficients
 - Probability outputs

Visuals required:

- Probability curve
- Bar chart comparing Yes vs No groups

9. Step 7 — Covariance

Calculate covariance for at least **three pairs**, such as:

- Distance & Duration
- Delay & Satisfaction
- AQI & Temperature

Visuals required:

- Scatter graphs
- Covariance table

Explain whether variables move together or independently.

10. Step 8 — Visualisation Portfolio

Create a final set of visuals:

- Bar chart
- Pie chart
- Line graph
- Scatter graph
- Heat map
- Infographic (PowerPoint or Canva allowed)
- 3D chart
- Optional: 3D map/model

11. Step 9 — Final Report

Write a **1,000–1,500 word** report including:

- Introduction
- Dataset description
- Methodology
- Findings from each statistical technique
- Visualisations
- Trends and patterns
- Recommendations for improving city mobility or air quality
- Conclusion